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Crop yield prediction using machine learning: An extensive and systematic literature review

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A R T I C L E I N F O

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A B S T R A C T

In recent years, agriculture has gained much attention regarding forecasting and prediction with the advancement of artificial intelligence techniques. Advancements in Machine Learning (ML) have significantly improved agricultural activities. In order to ensure food security and optimize resource allocation, precise crop yield prediction has become essential due to the growing global population and the effects of climate change on agricultural production. In this research, we conducted a Systematic Literature Review (SLR) to identify and synthesize techniques and attributes utilized in crop yield prediction research between the years of 2017 and 2024. This extensive search yielded 184 eligible papers from eight electronic sources. A total of 97 papers have been chosen for further analysis based on inclusion and exclusion criteria. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) technique has been employed to search, screen, and select relevant research papers, resulting in a comprehensive and unbiased review. According to this analysis, the most used features are temperature, soil type, and vegetation. Also, the most applied machine learning algorithms are Linear Regression (LR), Random Forest (RF), and Gradient Boosting Trees (GBT) whereas the most applied deep learning algorithms are Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM). An additional search has been performed in order to identify some hybrid ML-based models. As per the evaluation metrics, RMSE, R-square, and MAE are found to be the mostly favored. Eventually, this review offers valuable insights into the state-of-the-art algorithms in crop yield prediction and suggests future directions for researchers aiming to address existing difficulties and limitations.

1. Introduction

Agriculture's development is crucial for human civilization, providing food and enabling crop cultivation since communities formed. Farmers worldwide rely on it to meet local and global food needs, a method that has supported populations for generations and remains vital today. The estimate from the World Health Organization [111] indicates that 820 million people worldwide still have insufficient access to food. The Sustainable Development Goals set by the United Nations aim to eradicate hunger, achieve food security, and promote sustainable agriculture by 2030. [65]. The Food and Agriculture Organization (FAO) has projected a 60 percent increase in the demand for food to meet the needs of the projected global population of 9.3 billion by 2050 [104,65]. Hence, predicting crop yield can provide essential data needed to formulate a viable strategy to reach the goal of ending hunger [111].The crop yield is a commonly used measure that indicates the amount of agricultural production harvested per unit of land area. In traditional agriculture, farmers rely on their experience and some basic environmental factors such as rainfall, temperature, soil type, and pesticides used to predict crop yields. Utilizing previous record data based on these

Abbreviations: KNN, K-nearest neighbor; EL, Ensemble learning; RF, Random forest; Nn, Neural network; Ml, Machine learning; DL, Deep learning; Ai, Artificial intelligence; NDVI, Normalized difference vegetation index; LAI, Leaf area index; modis NDVI, Moderate resolution imaging spectroradiometer NDVI; EVI, Enhanced vegetation index; NMAE, Normalized mean absolute error; FLEM, Farm-level model error; RPD, Ratio of performance to deviation; RRMSE, Relative root mean square error; MAPE, Mean absolute percentage error; NRMSE, Normalized root mean square error; medae, Median absolute error; MSLE, Mean squared logarithmic error; RMSE, Root Mean Square Error.

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factors enables early crop prediction, helping farmers and agronomists make informed harvest decisions. Collecting data and interpreting it externally is necessary with this approach, which can be burdensome for farmers who may not completely understand the underlying concept [54]. Different forecasting methodologies have been created and assessed by researchers worldwide in the agricultural and related scientific fields [108].

1.1. Literature survey

Using machine learning (ML) for crop yield prediction can be challenging, particularly in selecting appropriate algorithms. Despite significant research, the growing volume of data presents opportunities for improvement. In this review article, we examine the current state of machine learning in crop yield prediction, drawing on a comprehensive search of the publications. Our analysis also includes an examination of other survey and review papers related to this field. The findings of these traditional survey and review papers are described and evaluated in this section. A Systematic Literature Review (SLR) was carried out by [106] in order to determine the features and algorithms utilized in crop yield prediction studies. Out of 567 relevant studies, 50 were selected for further analysis. The features that were used most are rainfall, temperature, and soil type, with Artificial Neural Networks being the most applied algorithm. The study also identified 30 deep learning-based studies, with Convolutional Neural Networks (CNN) being the most widely used deep learning algorithm. Other widely used algorithms include Deep Neural Networks (DNN) and Long-Short-Term Memory (LSTM). [55] researched various machine learning (ML) methods for predicting early grain yield at the field level. In the period from 2014 to 2021, 46 studies were examined, showing extensive data gathering, preprocessing, model training, and assessment. The findings demonstrate diverse prediction timeframes, extensive input data, limited use of feature selection techniques, and variation in the reporting of performance metrics.

Investigation on machine learning methods for precise prediction of crop yield as well as nitrogen status estimation was covered by [21]. The study concludes that quick developments in ML and sensing technologies will lead to more affordable ways to estimate crop and environmental conditions better. [97] examined the adoption of machine learning (ML)–based optical sensing for predicting biomass yield in managed grasslands. Results show a broad array of vegetation indices, focus on feature selection, low reporting rate, higher variability, and the need for greater comparability of study designs and results.

However, recent technological developments have allowed for the creation of a sophisticated deep-learning crop yield prediction model [65]. What sets it apart from other conventional machine learning techniques is its ability to analyze both unlabeled and unstructured data. In addition to examining how environmental variables and vegetation indices affect crop yield, [65] outlined the knowledge gaps that still exist in a specific area of deep learning methodologies. According to the study, the most popular deep learning techniques are Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM), with satellite remote sensing technologies like MODIS being the most used. Challenges include improving model accuracy, providing accurate information to agronomists, farmers, and policymakers, and addressing the black box property. [44] reviewed the development of deep learning-based object detection methods for crop counting, comparing them with other methods, and addressing issues like dataset limitations and occlusion in complex environments.

The implementation of machine learning and deep learning techniques has led to a noteworthy shift from crop-based farming to livestock farming. [34] reviewed recent ML use in precision livestock farming (PLF) for grazing and animal health, highlighting opportunities and challenges. Challenges include developing hybrid models, addressing grazing and health issues, and integrating soil and pasture variables. [60] conducted a systematic literature review that explored the use of deep learning in precision cattle farming, focusing on health and identification. Out of 678 studies, 56 are focused on cattle identification and health monitoring. 20 models are identified, with CNNs being the most popular. Nitrogen (N) is crucial for cereal crop growth and yield. Critical N dilution curves estimate N nutrition status. [59] compared N curves in wheat, maize, and rice, explained genotype differences, and discussed remote sensing methods for large-scale applications. [69] presented a systematic literature review on data analytics platforms in the agricultural sector, focusing on domains, stakeholders, objectives, technologies, data properties, and obstacles. Results show 33 features and 34 obstacles, indicating a popular topic in recent years Researchers have conducted studies on the potential application of IoT in agricultural practices, in addition to exploring the use of machine learning algorithms. [101] reviewed IoT applications in agro-industrial and environmental fields, identifying trends, architectures, and challenges, based on a systematic literature review from 2006 to 2016.

[71] examined AI-enabled Internet of Things applications in the management of poultry health and welfare, emphasizing robust monitoring systems and standardized parameters for disease prevention. It covered the opportunities and problems with AI and Internet of Things applications. Fruit growers rely on pre-harvest yield data for economic decisions and orchard management, but traditional methods are laborious. [42] analyzed research on automatic yield prediction using vegetation indices, machine learning, and image recognition, highlighting challenges and suggesting areas for improvement. In their review of the uses of LiDAR (Light Detection and Ranging) technology in precision agriculture, [83]covered trends and research challenges as well as metric estimation related to crops, digitization of trees and plants, object detection and navigation, planning, and decision support. [76] reviewed 29 studies that used UAV imagery to assess plant stand count. It was proposed that L*a*b* color space, deep learning, hyperspectral camera imagery, ideal flying height, and open-source platforms are necessary for a successful stand count. The review attempted to assist researchers, farmers, and producers in choosing and utilizing UAV-based algorithms for plant stand count estimation. The summary of the previous literature related to our field is presented in Table 1.

1.2. Motivation and contribution

The application of machine learning (ML) can be a challenging task, especially when it comes to selecting suitable algorithms for a particular task. Despite extensive studies on crop yield prediction, the ever-increasing volume of data in today's data-driven world leaves room for further improvement. To provide new researchers with a comprehensive understanding of prior work in this area, a systematic review has been conducted to present an overview of the previous research efforts. A Systematic Literature Review (SLR) identifies the potential areas in research that require further exploration within a specific problem domain while providing guidance to both practitioners and researchers intending to conduct new studies in that particular problem area [106]. This review aims to facilitate the identification of knowledge gaps and to provide a foundation for future research in this field.

Previous review articles have examined various elements of precision agriculture and crop forecasting. Nevertheless, a systematic literature review (SLR) needs to encompass the most significant studies to date. With an increasing number of papers published annually focused on crop yield forecasting and the development of algorithms for enhanced performance, existing review papers might not provide an all-encompassing overview of the most recent applied machine learning (ML) or deep learning (DL) frameworks such as. Consequently, such papers may be deficient in state-of-the-art methods for reliable crop yield predictions. Additionally, this paper emphasizes the use of hybrid models, alongside machine learning and deep learning, aiming to aid researchers in discovering and investigating better algorithms for superior outcomes. The growing prevalence of hybrid deep learning models is particularly noticeable throughout the study, resulting in

Table 1
Summary of Literature Survey.

Reference	Year	Summary	Contribution	Limitation
[97]	2023	Investigates the use of optical sensing based on machine learning to forecast the yield of biomass in managed grasslands.	Provides insights into current practices and limitations of ML-based optical sensing for biomass yield prediction.	Focuses on published articles between 2015 and 2022, potentially excluding newer research.
[55]	2023	Investigate Machine Learning (ML) approaches for early prediction of grain yield at the field scale.	Identifies several prediction horizons ranging from a few weeks to more than eight months before harvest and demonstrates the variety of input data that were utilized.	Hybrid models were not covered extensively.
[44]	2023	Reviewed the state of development for deep learning-based detection of objects and crop counting.	Introduces public datasets for crop counting algorithms and also provides an in-depth analysis of object detection for crop counting.	Discusses only two most widely used models for crop counting.
[83].	2023	Reviewed the application of LiDAR systems in precision agriculture, focusing on crop cultivation.	Presents a taxonomy of LiDAR applications in precision agriculture for example crop-related metric estimation.	Doesn't delve into specific challenges beyond mentioning "current trends and research challenges" without explicitly outlining them.
[76]	2022	Discusses the limitations of traditional methods for plant stand count and proposes UAV imagery with computer vision algorithms as a promising alternative	Provides the first synthesized information on using UAV imagery for plant stand count and identifies key factors affecting the accuracy of this method.	Limited to 29 reviewed studies and doesn't mention the potential challenges or errors associated with UAV-based plant stand count.
[42]	2022	Investigated and analyze research on automatic orchard yield prediction and estimation (2010–2021).	Reviews automatic yield monitoring approaches using multi-information systems and intelligent equipment and finds vegetation indices and machine learning as the most common tools for yield prediction.	Lacking exploration of novel applications or advancements.
[65]	2022	Reviewed research on using deep learning for crop yield prediction. It analyzes the advantages of deep learning, suitable remote sensing technologies, and factors influencing prediction.	Highlights satellite remote sensing (MODIS) as a popular technology and shows vegetation indices as the most used feature.	Specific food crop articles were not included, no comparison between ML and DL techniques.
[69]	2022	Systematically analyzed existing literature on data analytics platforms in agriculture. Identify key features, target stakeholders, objectives, utilized technologies, data properties, and encountered obstacles.	Conducts first-of-its-kind SLR focusing on data analytics platforms in agriculture and identifies 33 features and 34 obstacles associated with these platforms	Data is limited to studies published between 2010 and 2020.
[59]	2022	Reviewed research progress on the development of Nc curves for in-season crop nitrogen (N) diagnosis in wheat, maize, and rice.	Highlights how the Nc curve-based NNI can be used for seasonal crop N evaluation, crop N requirement estimation, and crop grain yield and quality expectation.	Limits the generalizability of the findings to other important crops with potentially different N dynamics and growth patterns.
[71]	2022	Reviewed research on AI-enabled Internet of Things (IoT) applications for poultry health and welfare management	Highlights the challenges of current poultry management and the potential of AI and IoT to improve welfare assessment, disease prevention, and overall poultry production.	The focus is specifically on poultry health and welfare, limiting the generalizability to other agricultural sectors and did not delve into the technical details of specific AI and IoT systems.
[60]	2021	Overviews current developments in deep learning applications in precision cattle farming with an emphasis on identification and health.	Analyzes 56 studies using deep learning for cattle health and identification also highlights Convolutional Neural Networks (CNNs) as the most common deep learning model used.	Challenges like cattle movement throughout tracking as well as real-time health monitoring in farm conditions were not addressed.
[106].	2020	Extraction and synthesis of features and algorithms from studies that predict crop yield	Identifies the most used features for machine learning and most widely used deep learning algorithms based on crop yield prediction.	Limited to an analysis of 50 studies and an additional 30 deep-learning studies.
[34]	2020	A systematic literature review with an emphasis on grazing and animal health about the application of ML to precision livestock farming.	Reviews current sensors, software, and data analysis techniques used in PLF (Precision Live Farming) with ML.	Potential limitations of ML in PLF that aren't mentioned
[21]	2018	Discusses advancements in machine learning-based methods for	Emphasizes the potential of combining ML with sensor data fusion and expert	The paper focuses on reviewing existing research and doesn't delve into the specific

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Table 1 (*continued*)

Reference	Year	Summary	Contribution	Limitation
		precise crop yield prediction as well as nitrogen status estimation that have been made during the past fifteen years of research.	knowledge for improved decision-making in precision agriculture.	limitations of the reviewed approaches.
[101]	2017	Reviewed the use of Internet of Things (IoT) in agro-industrial and environmental applications. It focuses on identifying application areas, trends, architectures, and challenges in these fields	Analyzes the usage of various sensors, actuators, communication technologies, and other components in IoT deployments for these fields.	The paper doesn't delve into specific details of the reviewed studies. The focus on a specific time frame (2006–2016) might limit the generalizability of the findings to the current state of the field.

strong and precise predictions. In addition to investigating hybrid models, this study also explored papers utilizing a range of widely recognized state-of-the-art models that are making waves in the field.

The main contributions of this manuscript are as follows –

To gather information about published research in the field of machine learning and crop yield prediction, studies have been examined from a variety of angles to gain insight. For this SLR study, four research questions (RQs) have been developed.

RQ1: What are the applied machine learning algorithms for crop yield prediction?
RQ2: Which features were used for experiments in previous studies?
RQ3: What evaluation metrics are used for model performance evaluation?
RQ4: What difficulties were faced by the researchers while conducting research?

The rest of the article is structured as follows: The approach is outlined in Section 2. The findings of the systematic literature review (SLR) are outlined in Section 3. The investigation into using deep learning to predict crop yields is detailed in Section 3. The analysis of this paper is provided in Section 4 and summarized in Section 5.

2. Materials and methods

2.1. Selected and data gathering procedure

Machine learning has been implemented in various fields including agriculture. Therefore, a systematic approach is necessary to collect relevant studies. Electronic databases such as Elsevier, ScienceDirect, MDPI, Google Scholar, and Scopus were utilized for collecting literature. The search keywords used were "crop yield prediction", "crop yield estimation", and "crop yield prediction using machine learning". These keywords helped filter the papers related to the field of interest. Subsequently, by analyzing the title and abstract, suitable papers were selected. The selection process can be viewed in the PRISMA flow diagram depicted in Fig. 1. A visual representation of the information flow in a systematic review is called a PRISMA flow diagram. The PRISMA flow diagram illustrates the overall research selection process [3]. To improve the quality and openness of stating in these kinds of studies, the PRISMA (Preferred Reporting Items for Systematic Reviews and

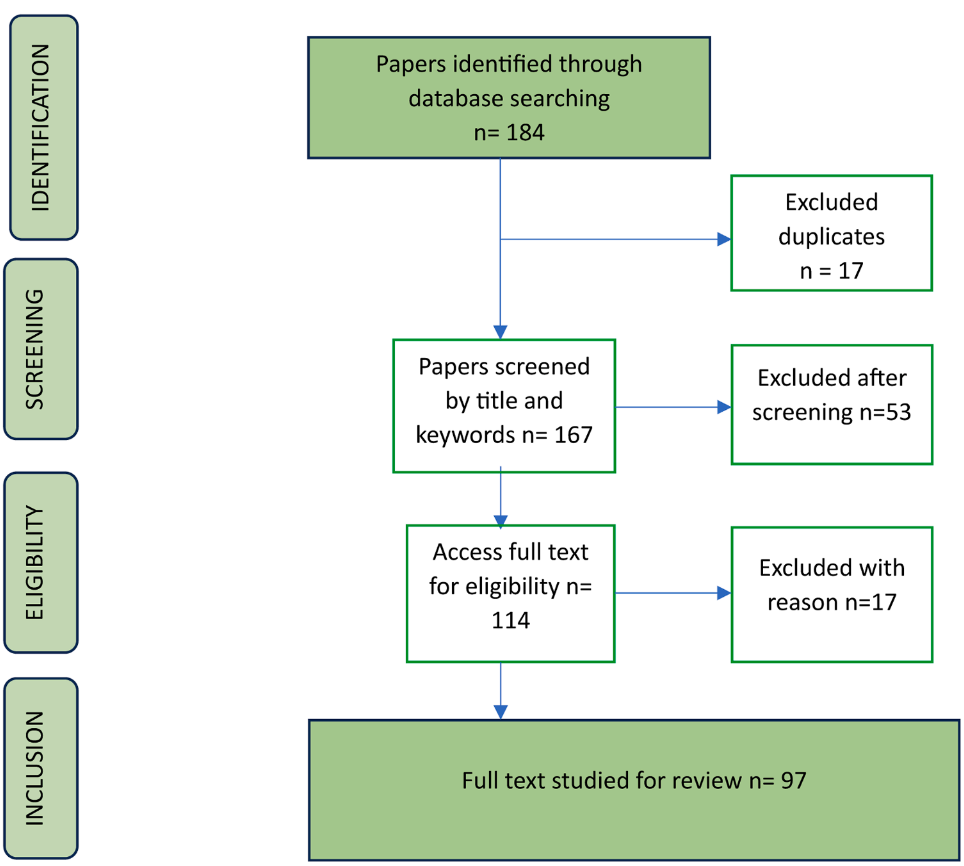

Fig. 1. PRISMA flow diagram.

Meta-Analyses) collection of reporting guidelines was developed. Usually, the flow diagram begins with the first number of studies found by database searches which is the first stage of identification. The next stage then lists the number of studies that are left after duplicates are eliminated, as well as the number of studies that are eliminated at each stage due to predetermined eligibility requirements. To determine the records' relevance regarding the research question or objective, this stage includes screening the abstracts and titles of the identified records. The flow diagram shows how many studies made it past the first round of screening and onto the next stage of eligibility. Full-text publications of possibly relevant research are currently evaluated in context with preset eligibility requirements. In the last inclusion stage, the total number of studies that are a part of the meta-analysis or systematic review is displayed. These studies are considered appropriate for additional analysis because they satisfy the predetermined eligibility requirements. The flow diagram shows how many studies were eliminated at each stage as well as the rationale behind those removals. Irrelevance to the research question, poor research methodology, insufficient data, or failure to satisfy particular inclusion criteria are common grounds for exclusion.

2.2. Search criteria

The strategy for searching articles has been thoughtfully developed to correspond with the specific research questions posed directly and to achieve the overarching goals of the systematic literature review. The search was conducted from eight different popular databases namely Science Direct, MDPI, IEEE Xplore, Springer, Elsevier, Google Scholar, Research Gate, and Scopus, and was limited to the English language. The logical operator that was employed to create the search string was AND. Following the initial application of the exclusion, i.e. not relatable to the agriculture factor, a more intricate search string was created to ensure that no pertinent studies were overlooked. This is the final search string that was used: ("artificial intelligence" or "machine learning" or "deep learning" and "data mining" and "yield prediction" or "yield forecasting" or "yield estimation"). When utilizing "machine learning" or "deep learning" as a standalone search term, researchers often encounter a vast array of published articles spanning multiple application fields. Many of these articles may not align closely with the specific objectives of the review, leading to an overwhelming amount of irrelevant information and complicating the search process. To address this issue, it is advisable to refine the search strategy by incorporating more targeted keywords. By employing a search string such as "crop yield prediction" AND "machine learning" AND "deep learning," the focus can be effectively narrowed. This approach significantly diminishes the likelihood of retrieving unrelated studies, thereby ensuring that the search results are more pertinent to the intended scope of the review. This strategic refinement is commonly practiced by researchers conducting systematic literature reviews focusing on machine learning and deep learning applications in smart farming [82,106,65]. Such an approach not only streamlines the research process but also enhances the relevancy of the gathered literature More details about the exclusion process are thoroughly discussed in the next part of this chapter.

2.3. Selection execution

The studies were examined and ranked according to the exclusion criteria in order to determine the requirements for the systematic review and eliminate irrelevant studies. These are the exclusion criteria (EC) presented as follows:

Exclusion Criteria 1: Paper that was not relevant to the agriculture sector.
Exclusion Criteria 2: Paper written in any other language except English.
Exclusion Criteria 3: Paper that has been already retrieved.
Exclusion Criteria 4: Paper is a review/survey journal.
Exclusion Criteria 5: Full paper is not available.
Exclusion Criteria 6: Paper was published before 2017.

After applying exclusion criteria 1, 2, and 6, a total of 184 papers were gathered through database searching, as shown in Fig. 1. Subsequently, 17 duplicate papers were removed based on exclusion criteria 3. Following this, 53 papers were excluded through title and keyword screening, leaving 114 papers. Further exclusions, such as the unavailability of the full paper, led to the removal of 17 more papers. Ultimately, after applying all exclusion criteria, a total of 97 papers remained for analysis.

In Table 2, it is evident that the majority of our papers were sourced from electronic databases such as Science Direct, MDPI, and IEEE Xplore. After applying the initial three exclusion criteria, it was found that most of the papers were unsuitable for the study. Table 2 shows the number of papers initially collected from each database. Subsequently, 78 papers were selected after all the aforementioned criteria were applied. To address the four research questions, all retrieved papers were thoroughly analyzed. After extracting all necessary data from the selected papers, those are presented in tabular format in the results section.

3. Results

Fig. 2 illustrates the distribution of selected publications by year. Papers published before 2017 have been excluded based on criterion number six. A significant increase can be observed in publications from 2020 onwards. The number of publications in 2017, 2018, and 2019 remained relatively low. However, there was a notable rise in 2020, with the trend continuing upward in 2021, 2022, and 2023. The increase in publications since 2020 can be largely attributed to a surge in interest in this specialized field of research. This is particularly evident in the area of agricultural technology, where the application of deep learning algorithms for predicting crop yields has become increasingly prominent in recent years. Researchers are recognizing the potential of these advanced computational techniques to analyze complex agricultural data, leading to more accurate forecasts. To enhance the understanding of applied algorithms, the research papers have been categorized based on the distinction between machine learning and deep learning algorithms. This classification provides a clearer view of the methodologies employed in prior studies, allowing for a comparison of their processes and outcomes. Such a systematic approach not only helps in elucidating the advancements made in the field but also highlights the evolving landscape of predictive analytics in agriculture.

The research conducted did not impose any exclusion criteria regarding the types of publications when searching through various databases. As a result, the scope of this study encompasses a wide range of publication types, including conference papers. The distribution of

Table 2
Distribution of papers across different databases showcasing initial and paper numbers after applying exclusion criteria.

Database	No. of paper initially retrieved	No. of paper after exclusion criteria	Papers Percentage (%)
Science Direct	73	27	28 %
MDPI	31	25	26 %
IEEE Xplore	44	14	14 %
Springer	34	11	11 %
Elsevier	60	9	9 %
Google Scholar	26	9	9 %
Research gate	15	2	2 %
Scopus	10	1	1 %
Total	292	97	100 %

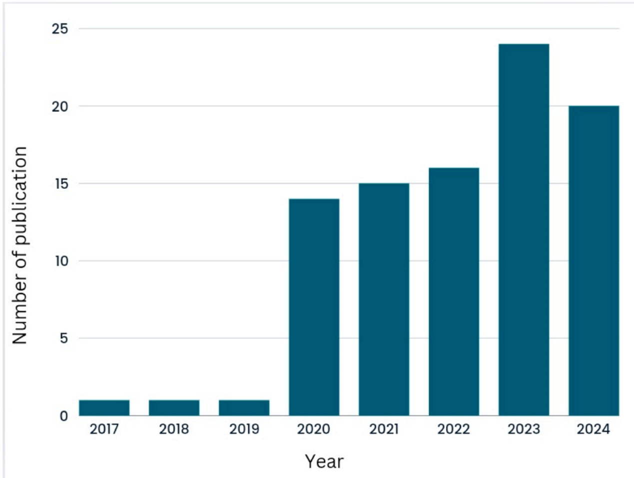

Fig. 2. Distribution of the selected publications per year from 2017 to 2024.

these different publication types is visually represented in the accompanying pie chart illustrated in Fig. 3. The chart reveals that a significant majority of the articles accessed for this study are journal articles, highlighting their prominence in the available literature. As illustrated, conference papers account for approximately 25 % of the total number of publications reviewed, while book chapters make up about 15 %. It is noteworthy that a substantial portion of these publications are open-access, facilitating greater accessibility for researchers and practitioners looking to engage with the findings presented in these works.

3.1. Machine learning algorithms for crop yield prediction

Table 3 below provides an overview of selected publications that utilize various machine-learning algorithms for crop yield prediction from 2019 to 2024. The Random Forest (RF) algorithm emerges as the most commonly used model throughout this period, highlighting its popularity and reliability in crop yield prediction tasks. Support Vector Regression (SVR) and Decision Trees (DT) have also been consistently applied from 2020 to 2023, demonstrating their ongoing relevance.Between 2022 and 2024, there has been an increase in the diversity of models used, including advanced ensemble methods such as XGBoost, Gradient Boosting, and Stacking Regression. In 2024, the landscape of machine learning for crop yield prediction has evolved to include more complex models like Interfused Machine Learning with Advanced Stacking Ensemble (IML-ASE) and multi-attribute weighted tree-based

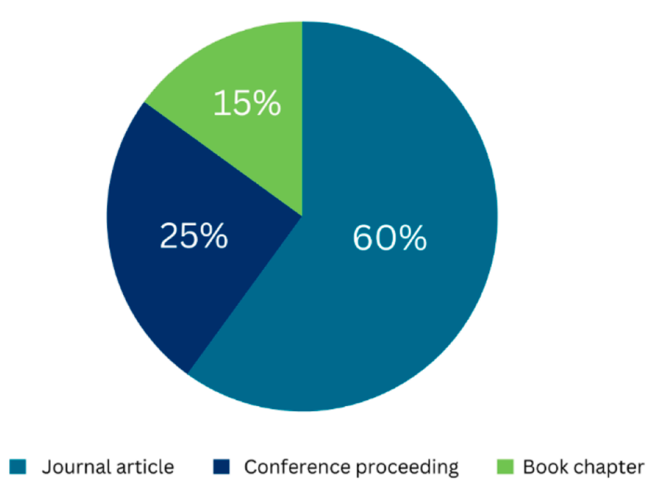

Fig. 3. The distribution of the types of collected publications and their corresponding percentages.

Table 3
A comprehensive collection of machine learning publications on crop yield predictions from 2019 to 2024, highlighting the algorithms used in each publication.

Retrieved from	Reference	Title	Algorithm	Year
Google Scholar	[72]	Performance Evaluation of Best Feature Subsets for Crop Yield Prediction Using Machine Learning Algorithms	ANN, SVR, KNN, and RF	2019
ResearchGate	[74]	Performance Evaluation of Machine Learning Techniques for Mustard Crop Yield Prediction from Soil Analysis	K-Nearest Neighbor (KNN), Naïve Bayes, Multinomial Logistic Regression, Artificial Neural Network (ANN), Random Forest.	2020
IEEE Xplore	[52]	Supervised machine Learning Approach for Crop Yield Prediction in Agriculture Sector	Random Forest, Decision Tree, SVR	2020
MDPI	[110]	Carrot Yield Mapping: A Precision Agriculture Approach Based on Machine Learning	Random Forest (RF) Regression	2020
MDPI	[1]	Crop Yield Prediction through Proximal Sensing and Machine Learning Algorithms	Linear Regression, Elastic Net, k-Nearest Neighbor, Support Vector Regression	2020
Elsevier	[38]	Integrated phenology and climate in rice yields prediction using machine learning methods	BP, SVM, RF	2021
Science Direct	[27]	Improved Yield Prediction of Ratoon Rice Using Unmanned Aerial Vehicle-Based Multi-Temporal Feature Method	Random Forest	2021
Elsevier	[75]	Analysis of agricultural crop yield prediction using statistical techniques of machine learning	Gradient Boosting Regressor, Random Forest Regressor, SVM and Decision Tree Regressor	2021
MDPI	[17]	Sugarcane Yield Mapping Using High-Resolution Imagery Data and Machine Learning Technique	Random Forest (RF) Regression, Multiple Linear Regression (MLR)	2021
Science Direct	[49]	Crop yield forecasting using data mining	Random Forest, XGBoost Classifier, KNN Classifier, Logistic Regression	2021
IEEE Xplore	[4]	Estimation of Crop Yield From Combined Optical and SAR Imagery Using Gaussian Kernel Regression	Gaussian Kernel Regression	2021
MDPI	[56]	Improving Potato Yield Prediction by Combining Cultivar Information and UAV Remote	Random Forest Regression (RFR), Support Vector Regression (SVR)	2021

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Table 3 (*continued*)

Retrieved from	Reference	Title	Algorithm	Year
		Sensing Data Using Machine Learning		
Google Scholar	[87]	Coupling machine learning and crop modeling improves crop yield prediction in the US Corn Belt	Ensemble of Linear regression, LASSO, LightGBM, Random Forest, XGBoost	2021
Elsevier	[77]	Machine learning for large-scale crop yield forecasting	Ridge Regression, K-nearest Neighbors Regression, Support Vector Machines Regression, Gradient Boosted, Decision Trees	2021
MDPI	[57]	Toward Automated Machine Learning-Based Hyperspectral Image Analysis in Crop Yield and Biomass Estimation	AutoML (Auto-Sklearn)	2022
IEEE Xplore	[85]	Ensemble Machine Learning Techniques Using Computer Simulation Data for Wild Blueberry Yield Prediction	Stacking Regression (SR), Cascading Regression (CR), Light Gradient Boosting Machine (LGBM), Gradient Boost Regression (GBR), XGBoost, Ridge	2022
Springer Link	[67]	Machine-Learning-Based Regional Yield Forecasting for Sugarcane Crop in Uttar Pradesh, India	RF, DT, LR	2022
Springer Link	[107]	Prediction of grain yield and nitrogen uptake by basmati rice through in-season proximal sensing with a canopy reflectance sensor	Regression Algorithm	2022
Elsevier	[19]	Crops yield prediction based on machine learning models: Case of West African countries	DT, multivariate logistic regression, and k-nearest neighbor models	2022
MDPI	[15]	Prediction of Field-Scale Wheat Yield Using Machine Learning Method and Multi-Spectral UAV Data	Gaussian Process Regression, Support Vector Machine Regression, Random Forest Regression	2022
MDPI	[64]	Machine Learning Applied to Tree Crop Yield Prediction Using Field Data and Satellite Imagery: A Case Study in a Citrus Orchard	Orthonormal Automatic Pursuit	2022
IEEE Xplore	[86]	Tackling Food Insecurity Using Remote Sensing and Machine Learning-Based Crop Yield Prediction	LASSO, RF, XGB	2023
MDPI	[53]	Crop Yield Prediction Using Machine Learning Models: Case of Irish Potato and Maize	Random Forest, Polynomial Regression, and Support Vector Regressor	2023
Science Direct	[16]	Improving crop yield prediction accuracy by embedding phenological heterogeneity into model parameter sets	Linear Regression (LR), Decision Tree (DT), Random Forest (RF)	2023
IEEE Xplore	[91]	Crop Yield Prediction: Robust Machine Learning Approaches for Precision Agriculture	RF, XGBoost, KNN, MLP, AdaBoost, DT	2023
Science Direct	[93]	Machine learning for yield prediction in Fergana valley, Central Asia	Linear Regression (LR), Decision Tree (DT), Random Forest (RF)	2023
Science Direct	[58]	Ensemble learning prediction of soybean yields in China based on meteorological data	K-nearest neighbor (KNN), Random Forest (RF), and Support Vector Regression (SVR)	2023
Google Scholar	[63]	Using machine learning for crop yield prediction in the past or the future	Regularized linear models, Random Forest, Artificial neural networks	2023
MDPI	[103]	Integrating Remote Sensing and Weather Variables for Mango Yield Prediction Using a Machine Learning Approach	Random Forest, Support Vector Regression, eXtreme Gradient Boosting (XGBOOST), RIDGE, LASSO, Partial Least Square Regression (PLSR)	2023
Science Direct	[109]	Prediction of winter wheat yield and dry matter in North China Plain using machine learning algorithms for optimal water and nitrogen application	Linear Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), Ensemble Learning (EL), Gaussian Process Regression (GPR)	2023
Springer Link	[98]	Forecasting carrot yield with optimal timing of Sentinel 2 image acquisition	Multivariate Model	2024
Science Direct	[80]	CropCast: Harvesting the future with interfused machine learning and advanced stacking ensemble for precise crop prediction	Interfused Machine Learning with Advanced Stacking Ensemble (IML-ASE)	2024
IEEE Xplore	[7]	Accurate Wheat Yield Prediction Using Machine Learning and Climate-NDVI Data Fusion	Support Vector Machine (SVM), Random Forest (RF), Least Absolute Shrinkage and Selection Operator (LASSO)	2024
Science Direct	[79]	Crop yield prediction using multi-attribute weighted tree-based support vector machine	MAWT-SVM, Genetic Algorithm	2024
Science Direct	[62]	Evaluating Machine Learning Models	Multiple Linear Regression,	2024

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Table 3 (*continued*)

Retrieved from	Reference	Title	Algorithm	Year
		and Identifying Key Factors Influencing Spatial Maize Yield Predictions in Data-Intensive Farm Management	Multilayer Perceptron, Decision Tree, Random Forest (RF)	
IEEE Xplore	[43]	Incorporating Meteorological Data and Pesticide Information to Forecast Crop Yields Using Machine Learning	Gradient Boosting K-Nearest Neighbors	2024
Springer Link	[26]	Sweet corn yield prediction using machine learning models and field-level data	Random Forest	2024
Springer Link	[33]	Machine learning approach for satellite-based subfield canola yield prediction using foral phenology metrics and soil parameters	Random Forest Regression (RFR)	2024
Science Direct	[115]	Within-season crop yield prediction by a multi-model ensemble with integrated data assimilation	Particle Filtering (PF) Algorithm	2024
Springer Link	[96]	Sugarcane yield estimation in Thailand at multiple scales using the integration of UAV and Sentinel-2 imagery	Random Forest Regression (RFR)	2024
MDPI	[37]	Comparative Evaluation of Remote Sensing Platforms for Almond Yield Prediction	Random Forest Regressor, XGBRegressor, Gradient Boosting regressor, Bagging Regressor, and AdaBoost Regressor	2024
Science Direct	[22]	The effect of dataset construction and data pre-processing on the eXtreme Gradient Boosting algorithm applied to head rice yield prediction in Australia	eXtreme Gradient Boosting (XGBoost	2024
Science Direct	[109]	Meta-knowledge guided Bayesian optimization framework for robust crop yield estimation	LightGBM (Gradient Boosting Framework) Bayesian Optimization	2024
Science Direct	[35]	Beyond assimilation of leaf area index: Leveraging additional spectral information using machine learning for site-specific soybean yield prediction	Gradient-boosted tree regressor (GBTR)	2024
Science Direct	[6]	Integrating Sentinel-1, Sentinel-2, and topographic data into soybean yield modeling	Random Forest Regression (RF), K Nearest Neighbor (KNN), Multiple Linear Regression	2024

Table 3 (*continued*)

Retrieved from	Reference	Title	Algorithm	Year
		using machine learning	(MLR), Decision Tree (DT)	
Springer Link	[112]	Enhancing direct-seeded rice yield prediction using UAV-derived features acquired during the reproductive phase	multiple linear regression (MLR), partial least squares regression, artificial neural networks, random forest regression, and support vector regression (SVR).	2024

Support Vector Machines (SVM). Overall, the trend from 2022 to 2024 indicates a shift towards more complex and ensemble-based models, such as XGBoost, Stacking Regression, and hybrid approaches.

In addressing the first research question (RQ1), an examination of the machine learning algorithms employed in the papers has been conducted and documented in Table 4. Linear Regression emerges as the most frequently utilized algorithm. Additionally, Random Forest (RF), Gradient Boosting Trees, SVM (Support Vector Machine), Decision Tree (DT), and k-Nearest Neighbor (k-NN) are also identified as prominently featured algorithms.

Research question two (RQ2) was addressed by listing all the features from the studied papers in Fig. 4. Features play a crucial role in crop yield prediction. Fig. 4 illustrates that the most commonly used features include Soil Type, Temperature, Vegetation, Yield Data, Normalized Vegetation Index (NDVI), Rainfall, Soil Properties, Imagery Data, Water Variable, and Sowing. These features were then grouped into 8 categories and presented in Fig. 5. For example, all soil variables such as soil type, soil pH, and soil texture were grouped into Soil Information. Crop Information included variables like crop types, crop weight, and crop growth stages. Weather Information grouped weather data such as temperature, rainfall, and precipitation. Field Management included field adjustment variables like irrigation data, plant spatial arrangement, and field sizes. Humidity grouped variables such as humidity, vapor, and vapor pressure. All types of image data were categorized under Imagery Data. Lastly, features that did not fit into any of the predefined categories were compiled into a distinct category labeled 'Other.' This category includes various indices such as the Normalized Difference Vegetation Index (NDVI), the MODIS-derived NDVI, and the Enhanced Vegetation Index (EVI). Each of these indices serves a specific purpose in monitoring vegetation health and land surface conditions, thus warranting their inclusion in a separate grouping for clarity and organization.

The most commonly used evaluation metrics for addressing research question (RQ3) are RMSE, R2, MAE, and MSE, as shown in Table 5. All evaluation metrics used in the collected publications have been identified and presented in the table, along with the number of times each metric has been used. In order to address research question four (RQ4) and to identify the challenges faced by researchers, the papers were thoroughly studied, especially focusing on the limitations. It was found

Table 4
Top Machine Learning Algorithms Used for Crop Yield Prediction.

Most used machine learning algorithms	No. of Time Used
Linear Regression	37
Random Forest (RF)	34
Gradient Boosting Trees	25
SVM (Support Vector Machine)	24
Decision Tree (DT)	15
k-Nearest Neighbor (k-NN)	12
Neural Network	9
Ensemble Learning	8
Logistic Regression	3

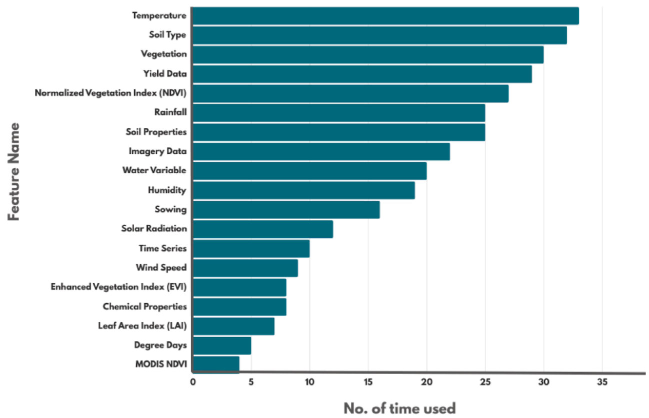

Fig. 4. Visualization of Mostly Used Features for Crop Yield Prediction.

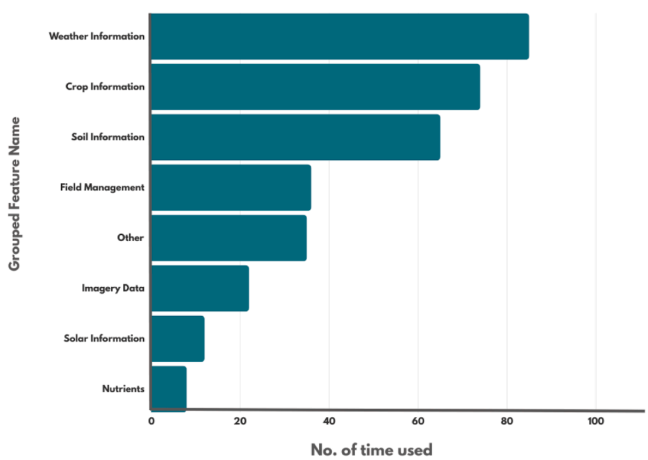

Fig. 5. Frequency of Usage for Top Grouped Features Categorized by Similarities.

Table 5
Assessment of evaluation metrics according to their implementation in analyzed publications.

Evaluation Metric	Used Number of Time
Root Mean Square Error (RMSE)	48
R-squared (R^2)	35
Mean Absolute Error (MAE)	31
Mean Squared Error (MSE)	12
Relative Root Mean Square Error (rRMSE)	6
Mean Absolute Percentage Error (MAPE)	6
Normalized Root Mean Square Error (NRMSE)	5
Median Absolute Error (MedAE)	2
Mean Squared Logarithmic Error (MSLE)	2
Explained Variance Score (Exp. Var.)	2
F-score	3
Accuracy	9
Precision	3
Recall	3
Specificity	3
Bias	2
Dice Coefficient	1
Lin's Concordance Correlation Coefficient (LCCC)	2
Normalized Mean Absolute Error (NMAE)	1
Farm-Level Model Error (FLEM)	1
RPD (Ratio of Performance to Deviation)	1

that authors often encountered challenges due to insufficient datasets. This lack of data included variations in climatic conditions, different types of vegetation, and longer time-series of yield data. Some studies suggested that by increasing the size of the dataset and incorporating crop physiological parameters, crop predictions could be made on a broader scale. In addition, a few papers mentioned the need for improvements, stating that weather information alone is not sufficient and that machine learning models require more hydrological inputs to make more accurate yield predictions.

3.2. Deep learning-based frameworks for crop yield prediction

Table 6 presents various deep learning-based models for crop yield prediction, highlighting the algorithms used, publication years, and sources for each study from 2019 to 2024. The most frequently used models are Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) networks, which appear multiple times across different years. In the earlier years, specifically from 2019 to 2021, DNNs and Recurrent Neural Networks (RNN) were the predominant models. However, recent studies from 2023 to 2024 have begun to emphasize the integration of Convolutional Neural Networks (CNN) with other models such as LSTM and Gated Recurrent Units (GRU). Furthermore, research papers from 2020 onward indicate an increasing trend in the use of UAV-based imagery and remote sensing data in combination with deep learning models for crop yield prediction, reflecting a heightened interest in utilizing spatial data for more accurate forecasts.

3.3. Hybrid and state-of-the-art model-based crop yield prediction

Table 7 presents a compilation of hybrid and state-of-the-art models used for crop yield prediction, detailing the algorithms employed, the year of publication, and the sources of each study. The most commonly utilized state-of-the-art models in these studies are Random Forest (RF), Long Short-Term Memory (LSTM) networks, and Support Vector Machine (SVM). These models consistently appear across multiple years, demonstrating their reliability in crop yield forecasting. These models are considered state-of-the-art due to their key characteristics. They use innovative techniques to solve complex problems and adapt to new data. Their strong generalization allows them to excel on both training and unseen data, making them versatile in real-world applications. Additionally, they consistently outperform competing models in various benchmarks. Their efficient computation ensures they are powerful yet resource-effective, enabling faster processing and lower operational costs. These attributes make them a top choice in the field.

Between 2020 and 2021, studies often combined Recurrent Neural Networks (RNN), LSTM, and machine learning techniques such as Support Vector Regression (SVR) and Random Forest. More recent studies from 2023 to 2024 have shifted towards using ensemble learning, fuzzy hybrid models, and multimodal data fusion to enhance prediction accuracy and robustness. Publications from 2022 to 2024 reveal a significant trend toward integrating deep learning with traditional machine learning algorithms like Gradient Boosting and Random Forest. This hybrid approach, which combines deep learning with algorithms such as SVM and XGBoost, has gained substantial attention. Additionally, there is a growing use of UAV-based data and remote sensing technologies to improve yield prediction accuracy. Recent models also integrate climate data, irrigation schedules, and various environmental variables for enhanced precision. This holistic approach, observed from 2021 to 2024, considers external factors influencing crop yield outcomes.

Table 8 provides an overview of the deep learning algorithms employed in the reviewed publications. The prominent deep learning algorithms utilized include CNN, LSTM, RNN, and DNN. Additionally, notable usage of Multi-Layer Perceptron (MLP), Bidirectional LSTM (Bi-LSTM), and DLMLP (Deep Learning Multi-Layer Perceptron) is evident, as depicted in Table 6. Brief descriptions of these algorithms are provided below.

Table 6
Deep Learning publications (2019–2024) on crop yield prediction focused on associated algorithms.

Retrieved from	Reference	Title	Algorithm	Year
Google Scholar	[50]	Crop Yield Prediction Using Deep Neural Networks	DNN	2019
IEEE Xplore	[5]	Crop Yield Prediction Using Deep Reinforcement Learning Model for Sustainable Agrarian Applications	Deep Recurrent Q-Network	2020
ResearchGate	[113]	Real-time Crop Classification Using Edge Computing and Deep Learning	SegNet, FCN-AlexNet	2020
IEEE Xplore	[100]	Multilevel Deep Learning Network for County-Level Corn Yield Estimation in the U. S. Corn Belt	RNN, CNN	2020
MDPI	[66]	Crop Yield Prediction Using Multitemporal UAV Data and Spatio-Temporal Deep Learning Models	CNN, LSTM	2020
MDPI	[36]	Deep Learning Based Prediction on Greenhouse Crop Yield Combined TCN and RNN	Temporal Convolutional Network, Recurrent Neural Network, LSTM	2021
Science Direct	[84]	Field-scale crop yield prediction using multi-temporal WorldView-3 and PlanetScope satellite data and deep learning	2D and 3D ResNet-18	2021
Science Direct	[105]	A deep learning multi-layer perceptron and remote sensing approach for soil health based crop yield estimation	Utilized Deep Learning Multi-Layer Perceptron (DLMLP) neural networks	2022
Science Direct	[78]	Pixel-based yield mapping and prediction from Sentinel-2 using spectral indices and neural networks	Recurrent Neural Network (RNN)	2023
MDPI	[12]	Improved Optimization Algorithm in LSTM to Predict Crop Yield	LSTM, CNN, RNN	2023
Springer Link	[11]	End-to-end 3D CNN for plot-scale soybean yield prediction using multitemporal UAV-based RGB images	3D CNN (DenseNet169, VGG-13, ResNet50)	2024
Springer Link	[32]	Prediction of cotton yield based on soil texture, weather conditions and UAV imagery using deep learning	CNN, GRU (Gated Recurrent Unit) Network	2024
Elsevier	[41]	A graph-based deep learning framework for field scale wheat yield estimation	Seq_Gra_Gd model	2024
Science Direct	[99]	Crop yield prediction using effective deep learning and dimensionality reduction approaches for Indian regional crops	WTDCNN (Wavelet Transform Deep Convolutional Neural Network)	2024
Science Direct	[25]	A deep learning framework for prediction of crop yield in Australia under the impact of climate change	Deep Neural Networks (DNN)	2024

- **Recurrent Neural Network (RNN):** An RNN's hidden state is determined by the inputs at the current time step as well as the hidden state from the time step prior. This serves as the network's "memory," storing data about the content that has already been viewed. The RNN updates its hidden state for every input in the sequence, which enables it to extract context from the sequence's past. Unlike other neural network types, an RNN can process sequences of variable lengths as a result of this feature. RNN has shown a low error rate for crop yield prediction.
- **Convolutional Neural Network:** CNNs are widely used in various applications due to their flexibility and ability to automatically identify hierarchical features from visual input. A typical CNN model consists of several convolutional and pooling layers, as well as a small number of fully connected (FC) layers. The design of CNNs involves several parameters, such as the number of filters, filter size, type of padding, and stride [51]. There are two methods to implement an architecture: using pre-defined architectures or starting from scratch. The first method's drawback is that it requires a lot of time, but the second approach offers time savings and superior performance.
- **Long short-term memory:** Recurrent neural networks (RNNs) frequently utilize the Long Short-Term Memory (LSTM) architecture for sequence modeling applications. The LSTM architecture is characterized by a more intricate structure that includes extra memory cells and gates, enabling it to selectively retain or discard information from preceding time steps. By taking into account past historical values, LSTM examines current unknown patterns and adapts itself based on the complete patterns, enabling it to make future predictions [13].
- **Deep Neural Network:** A DNN is made up of follow-up layers of neurons that lead to an output layer. These layers can be viewed as sequential representations of the data being entered. The DNN architecture uses neurons as elementary computing units with an activation function to model layers connected using weighted links. During the training phase, the architecture evaluates weight vectors and uses forward and backward passes to compute prediction errors and gradients.
- **Hybrid and State-of-the-art Models:** Several researchers have attempted to combine deep learning algorithms and machine learning algorithms, as well as combining multiple deep learning algorithms. The yield estimation framework presented in [114] estimates rice yields using an unmanned aerial vehicle equipped with a multispectral camera. The framework selects important vegetation indices and generates a dataset for data augmentation using the deep neural network (DNN) and synthetic minority oversampling technique (SMOTE). DNN is used to train the model, and alternative approaches are contrasted. In addition to offering a fresh method for

Table 7
List of hybrid and state-of-the-art models for crop yield prediction, covering the years 2020 to 2024.

Retrieved from	Reference	Title	Algorithm	Year
Google Scholar	[68]	Hybrid Machine Learning Models for Crop Yield Prediction	ANN-ICA, ANN-GWO	2020
MDPI	[29]	A Hybrid CFS Filter and RF-RFE Wrapper-Based Feature Extraction for Enhanced Agricultural Crop Yield Prediction Modeling	RF, RF-RFE, CFS Feature Selection	2020
MDPI	[23]	Rice Crop Detection Using LSTM, Bi-LSTM, and Machine Learning Models from Sentinel-1 Time Series	RNN, LSTM, Bi-LSTM, SVM, RF, KNN, NB	2020
IEEE Xplore	[28]	Crop Yield Prediction Using Deep Reinforcement Learning Model for Sustainable Agrarian Applications	Deep Recurrent Q-Network, Deep LSTM Network, ANN, Gradient Boosting, RF, Bernoulli Deep Belief Network, Bayesian Artificial Neural Networks, Rough Auto Encoders, Interval Deep Generative Artificial Neural Networks	2020
Google Scholar	[2]	A Hybrid Approach for Crop Yield Prediction Using Machine Learning and Deep Learning Algorithms	Random Forest, Decision Tree, ANN, SVM, LSTM, RNN	2021
MDPI	[5]	Crop Yield Estimation Using Deep Learning Based on Climate Big Data and Irrigation Scheduling	Bidirectional Long Short-Term Memory, Long Short-Term Memory, Gated Recurrent Units, Bidirectional GRU, Convolutional Neural Network, Multi-Layer Perceptrons, Random Forest Regression	2021
Elsevier	[18]	Wheat yield predictions at a county and field scale with deep learning, machine learning, and google earth engine	Random forest, including DNN (deep neural networks), 1D-CNN (1D convolutional neural networks), and LSTM (long short-term memory networks)	2021
Scopus	[8]	Deep Learning Based Wheat Crop Yield Prediction Model in Punjab Region of North India	RNN-LSTM, Multivariate Linear Regression, ANN, Random Forest	2021
MDPI	[30]	Farm-Scale Crop Yield Prediction from Multi-Temporal Data Using Deep Hybrid Neural Networks	Hybrid Neural Network: CNN and RNN	2021
Google Scholar	[87]	Coupling machine learning and crop modeling improves crop yield prediction in the US Corn Belt	Linear Regression, LASSO, LightGBM, Random Forest, XGBoost, Ensemble models	2021
Google Scholar	[92]	Crop yield prediction integrating genotype and weather variables using deep learning	LSTM, Support Vector Regression with Radial Basis Function kernel (SVR-RBF) LASSO Regression	2021
Science Direct	[102]	Combining Machine Learning, Space-Time Cloud Restoration and Phenology for Farm-Level Wheat Yield Prediction	RF, GLM, SVM, KNN, Neural Networks, RPART	2021
MDPI	[9]	A Hybrid Approach to Tea Crop Yield Prediction Using Simulation Models and Machine Learning	AquaCrop simulation model, XGBoost regression	2022
Google Scholar	[70]	Hybrid Deep Learning-based Models for Crop Yield Prediction	CNN-DNN, CNN-XGBoost, CNN-RNN, CNN-LSTM, XGBoost	2022
Springer Link	[31]	UAV-based multi-sensor data fusion and machine learning algorithm for yield prediction in wheat	Cubist, SVM, DNN, RF, Ensemble Learning (Stacked RR)	2022
Springer Link	[73]	Early yield prediction in different grapevine varieties using computer vision and machine learning	SegNet, Support Vector Regression (SVR	2022
Science Direct	[116]	Early season prediction of within-field crop yield variability by assimilating CubeSat data into a crop model	Linear regression with APSIM and particle filter integration	2022
MDPI	[9]	A Hybrid Approach to Tea Crop Yield Prediction Using Simulation Models and Machine Learning	FAO AquaCrop Simulation Model Regression Algorithms	2022
MDPI	[114]	A Hybrid Synthetic Minority Oversampling Technique and Deep Neural Network Framework for Improving Rice Yield Estimation in an Open Environment	SMOTE, DNN	2023
MDPI	[45]	Automated Estimation of Crop Yield Using Artificial Intelligence and Remote Sensing Technologies	Fuzzy Hybrid Ensemble Classification, Gradient Boosting, Decision Tree, Multivariate Regressor, Random Forest, MobilenetV2	2023
MDPI	[24]	Crop Yield Prediction Using Hybrid Machine Learning Approach: A Case Study of Lentil (Lens culinaris Medik.)	RF, SVM, KNN, ANN	2023
Science Direct	[46]	Crop yield prediction using machine learning techniques	Multiple Linear Regression (MLR), Decision Tree Regression, Gradient Boosting, Elastic Net, Lasso, Feature engineering-based LSTM	2023

(continued on next page)

Table 7 (*continued*)

Retrieved from	Reference	Title	Algorithm	Year
Elsevier	[39]	Comparison of different machine learning algorithms for predicting maize grain yield using UAV-based hyperspectral images	Random Forest Regression, Light Gradient Boosting Machine, Convolutional Neural Network	2023
IEEE Xplore	[81]	A Multimodal Data Fusion and Deep Neural Networks Based Technique for Tea Yield Estimation in Pakistan Using Satellite Imagery	DT, MLP, SVR, Gaussian Process Regression (GPR), Multiple Linear Regression, RF, Gradient Boosting, XgBoost, Proposed Deep Neural Network(DNN)	2023
IEEE Xplore	[88]	Predicting Agriculture Yields Based on Machine Learning Using Regression and Deep Learning	Decision Tree, RF, XGBoost Regression, LSTM	2023
Elsevier	[47]	Crop Yield Prediction using Machine Learning and Deep Learning Techniques	Random Forest, Support Vector Machine (SVM), Gradient Descent, Long Short-Term Memory (LSTM), Lasso Regression	2023
IEEE	[61]	An Approach for Crop Prediction in Agriculture: Integrating Genetic Algorithms and Machine Learning	Random Forest Classifier, optimized with Genetic Algorithms (GAs)	2024
Science Direct	[90]	UAV remote sensing phenotyping of wheat collection for response to water stress and yield prediction using machine learning	Linear Regression, Support Vector Machine, Random Forest, Deep Learning H2O-3	2024
MDPI	[20]	Hybrid Deep Neural Networks with Multi-Tasking for Rice Yield Prediction Using Remote Sensing Data	Hybrid Deep Neural Network with Multi-Task Learning (MTL)	2024
Science Direct	[94]	Bridging the Gap between Crop Breeding and GeoAI: Soybean Yield Prediction from Multispectral UAV Images with Transfer Learning	Random Forest Regressor (RF), Gradient Boosting Regression (GB), Deep Neural Networks (DNN)	2024

Table 8
Usage frequency of Deep Learning algorithms for yield prediction, determined from analyzed publications.

Algorithm	No. of Time Used
CNN	13
LSTM	13
RNN	12
DNN	8
SegNet	2
Seq_Gra_Gd model	1
2D and 3D ResNet-18	1
FCN-AlexNet	1

other crops, the framework greatly increases the accuracy of yield estimation, making it appropriate for estimating rice yield. In order to effectively model and forecast complex phenomena like crop yield, [24] proposes a hybrid approach that combines feature selection and machine learning algorithms. The performance of the MARS-ANN and MARS-SVR hybrid frameworks is compared using fit model-building statistics. Because of their improved feature extraction and nonlinear adaptive learning capabilities, the MARS-based models performed better than individual models. The hybrid models exhibit better generalization and model-building capabilities. [10] used machine learning techniques and the AquaCrop simulation model from the Food and Agriculture Organization to compare methods for predicting tea yield. The National Tea and High-Value Crop Research Institute of Pakistan provided the data used, which covered the years 2016 through 2019. Weather, soil, crop, and agro-management data were used to calibrate the AquaCrop model. Compared to the simulation model, the machine learning regression algorithm outperformed it in yield prediction with fewer data. By integrating various data sources with a crop simulation model and machine learning algorithms, this study offers a method to enhance tea yield prediction.

When discussing state-of-the-art models, [48] combined the RF, SVM, Gradient Descent, LSTM, and Lasso Regression techniques to predict crop yield. They found that RF performed the best among the other algorithms. Upon analyzing Table 7, it was found that the most applied machine learning algorithms along with deep learning are RF, SVR, SVM, MLP, and Multiple Linear Regression (MLR). [40] attempted to compare the prediction outcomes using five deep learning methods of CNN, with varying numbers of convolutional kernels and five widely used machine learning methods (BP, RF, SVM, PLSR, and LightGBM). To estimate agricultural yield. [89] used deep learning methods such as CNN, and LSTM in addition to machine learning methods like decision trees, random forests, and XGBoost regression. To enhance the model's performance, researchers also combined multiple deep learning algorithms together. [14] found that their proposed IOF (Improved Optimizer Function) with LSTM can outperform CNN, RNN, and LSTM in crop yield prediction. It is evident from the analysis researchers are presently engaged in the exploration of a diverse array of machine learning and deep learning algorithms, encompassing both state-of-the-art models and innovative hybrid approaches., which are making substantial contributions to the field of precision agriculture.

In addition to the main content, a supplementary file has been included that features two detailed tables. The first table provides a thorough breakdown of the 21 distinct features analyzed across the various publications, offering insights into their specific characteristics and functionalities. The second table presents an overview of the evaluation parameters employed in these studies, detailing the criteria and methods used to assess each feature. This comprehensive information is designed to enhance understanding and facilitate further analysis of the data presented.

4. Discussion

Previously, many researchers have attempted to conduct a review survey on this topic. This kind of research can be influenced by validity threats, which may come in the form of external threats, construct validity threats, and reliability threats [95]. The broad search string of this study addressed the external and construct validity, covering the entire scope of this SLR. The procedures of this SLR have been clearly defined through the PRISMA flow diagram to ensure reliability. The process of conducting the previous surveys might be similar to some other work, but an extension of recent publications was necessary. In comparison to a similar study by [106], which reviewed 50 papers and addressed several important research questions, this current study aims to provide a more comprehensive analysis. Given that their work was published four years ago, it is essential for review surveys to incorporate more recent publications, as research in this field continues to grow each year. To address this gap, this paper expands the scope of the review to include 100 papers related to crop yield prediction. The applications of machine learning and deep learning algorithms are evaluated and the availability of data also increased over time. This played a role in changing the number of algorithms, evaluation parameters, features listed in the previous review survey, and the viewpoint of new researchers. Furthermore, this paper highlights the application of hybrid models, as well as machine learning and deep learning, which will assist researchers in understanding and exploring better algorithms for improved results The increasing adoption of hybridized deep learning models is notably evident throughout the study, yielding robust and accurate predictions. In addition to hybrid models, this study also identified several popular state-of-the-art models in the literature, such as Random Forest (RF), Long Short-Term Memory (LSTM) networks, and Support Vector Machines (SVM).

To further emphasize the significance of this work, another comparison with related research can be made. While [65] study focuses on the role of remote sensing technologies and the challenges associated with deep learning, this current study adopts a quantitative approach, providing broader insights into state-of-the-art algorithms and their limitations. Additionally, it incorporates a wider range of features, including temperature and soil type, alongside vegetation indices. Both studies contribute uniquely to the field: the former is more specialized, concentrating on deep learning and remote sensing, whereas the current study offers a comprehensive perspective on advancements in machine learning for crop yield prediction.

Despite the variations observed in individual studies, the overarching conclusions drawn from this systematic literature review (SLR) appear to maintain a consistent alignment with previous research. Convolutional Neural Networks (CNN) remain the most popular deep learning utilized for yield forecasting in the research community. Furthermore, the Root Mean Square Error (RMSE) continues to be recognized as the primary evaluation metric for assessing performance in various analytical contexts. This consistency suggests that the key themes and insights identified in the current analysis are likely reflective of broader trends within the field. In the field of machine learning, Linear Regression has emerged as the most frequently utilized algorithm, surpassing the popularity of Neural Networks. This shift can be attributed to several factors, including the simplicity of Linear Regression, its ease of interpretation, and the ability to provide quick and effective solutions for a range of regression problems. While Neural Networks are powerful tools capable of handling complex data patterns, Linear Regression remains a preferred choice for many practitioners, especially in situations where the relationship between variables can be adequately described with a linear equation. As compared to the study [106], this trend reflects the growing preference for techniques that balance performance with interpretability and computational efficiency.

Even though the initial search string aimed to cover crop yield prediction, the query returned thousands of publications. Among these publications, not all were relevant to the research topic. Since only selected papers were studied, it is possible that valuable research papers were not included in the survey. While studying papers, it was noticed that not all authors were clear about the required information for the survey, such as evaluation metrics. Details of the features and algorithms used were also not stated in some papers, leading to the exclusion of those papers from the research. Overall, the selected number of papers was suitable for extracting the required data to answer the research question of this present SLR.

To answer RQ1 this study examines the machine learning algorithms used in various papers, with Linear Regression being the most frequently used algorithm. Other prominent algorithms include Random Forest (RF), Gradient Boosting Trees, Support Vector Machine (SVM), Decision Tree (DT), and k-nearest Neighbor (k-NN). When it comes to the most commonly used machine learning algorithms, it's clear that Linear Regression is still widely used for predicting crop yield. However, just because it's used in many articles doesn't necessarily mean it's the best-performing algorithm. Other algorithms also appear to outperform Linear Regression when implemented alongside it. While studying deep learning-based research, it was noticed that deep learning plays a dominant role in crop yield prediction. Many researchers are trying to use deep learning algorithms for this purpose. Some of these algorithms, such as LSTM and CNN, have shown promising results and outperformed traditional machine learning algorithms in prediction. It can be said that deep learning will play a much larger role in this field in the future due to its superior performance and automatic feature extraction.

Features play a crucial role in crop yield prediction, with soil type, temperature, vegetation, and yield data being the most commonly used features**, answering RQ2**. In addition to the most commonly used features, there were also several popular features such as the Normalized Vegetation Index (NDVI), rainfall, soil properties, imagery data, water variables, and nutrients. These features are grouped into eight categories to provide an overall understanding of the most frequently used feature types. While grouping the features, no loss of features was ensured.

As for RQ3, the most commonly used evaluation metrics are RMSE, R2, MAE, and MSE. Most of the papers reviewed used at least one of these evaluation metrics for their model evaluation. Apart from these metrics, some papers also used evaluation metrics that were slightly different from the most commonly used ones. These include Relative Root Mean Square Error (rRMSE), Mean Absolute Percentage Error (MAPE), Normalized Root Mean Square Error (NRMSE), Median Absolute Error (MedAE), Mean Squared Logarithmic Error (MSLE), and Explained Variance Score (Exp. Var.). These metrics also appear to yield favorable results when evaluating the models.

The papers were thoroughly studied to identify challenges faced by researchers while attempting research on this field **while addressing RQ4.** Most of the challenges include insufficient datasets, variations in climatic conditions, different types of vegetation, and longer time series of yield data. Some studies suggest that by increasing the size of the dataset and incorporating crop physiological parameters, crop predictions could be made on a broader scale. Some papers did not mention difficulties and challenges, so some of them may still remain unidentified. However, another challenge remains in implementing the models in agricultural systems. The usefulness and accuracy of the models can only be justified by the farmers.

5. Conclusion

The paper explores various publications on crop yield prediction using different machine learning and deep learning algorithms. Additionally, the research emphasizes the significance of state-of-the-art and hybrid models, which combine the strengths of both machine learning and deep learning algorithms. Although several algorithms are frequently used in machine learning, none can be definitively labeled as the best performer. The most commonly utilized algorithms include Linear Regression, Random Forest, Gradient Boosting Trees, and Support Vector Machines (SVM). Many studies have employed multiple machine learning algorithms to assess and identify the best performer. Not only did the algorithms vary in the papers, but the dataset, features, scales, geographical position, and crops also differed. Some papers focused on predicting specific crops, while others concentrated on groups of crops. It was also noted that using a large number of features does not guarantee the best outcome. Selecting relevant features appears to be crucial in research. It can be concluded that for an accurate crop yield prediction model, it should be trained with a comprehensive dataset containing a variety of features. The popularity of deep learning in precision agriculture has surged, especially in the area of predicting crop yields. The study highlights several commonly used algorithms for this purpose, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Recurrent Neural Networks (RNN), and Deep Neural Networks (DNN). It's clear that in recent years, there has been a growing interest in exploring the applications of deep learning algorithms in predicting crop yields. Furthermore, several prominent state-of-the-art models, including Random Forest (RF), Long Short-Term Memory (LSTM) networks, and Support Vector Machines (SVM), are demonstrating expected outcomes and are popular among researchers. Researchers have found that combining different algorithms in various ways is essential for achieving the best results and comparing model performance. This systematic literature review serves as a significant resource for scholars, agronomists, and decision-makers working in agricultural planning and policy-making, portraying a thorough grasp of how machine learning can be leveraged to predict crop yields more precisely and reliably.

While this study did not extensively focus on the performance analysis of ML and DL models, the performance metrics from the work covered demonstrated that DL and Hybrid Architecture based models outperform others in evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). Furthermore, accurate crop yield prediction is intricately linked to the careful selection of both environmental and crop-specific attributes. Thus, achieving high predictive accuracy necessitates not only the adoption of robust model architectures but also the meticulous selection of relevant attributes. A notable limitation of this study is the lack of a detailed comparative analysis of individual model performances. Future research can address this gap by conducting a comprehensive evaluation of various ML and DL models, and benchmarking them across diverse datasets and metrics. Such an analysis in future work could provide deeper insights into the strengths and weaknesses of each model in specific scenarios.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Sarowar Morshed Shawon: Writing – original draft, Methodology, Investigation, Formal analysis. **Falguny Barua Ema:** Writing – original draft, Investigation, Formal analysis. **Asura Khanom Mahi:** Writing – original draft, Methodology, Investigation, Formal analysis. **Fahima Lokman Niha:** Writing – original draft, Methodology, Investigation, Formal analysis. **H.T. Zubair:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2024.100718.

Data availability

No data was used for the research described in the article.

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